SPACEX FALCON 9

First-Stage Landing Prediction

End-to-end data science capstone report

Data collection → wrangling → EDA → SQL → Folium → machine learning

Source: supplied IBM Skills Network / Coursera lab notebooks

Executive Summary

A multi-method analytical story around Falcon 9 first-stage landing success.

90	66.7%	18	15/18	0.877
ML/EDA records	landing-success label rate	test samples	correct test predictions	best 10-fold CV

- Data was collected from SpaceX APIs and a fixed Wikipedia snapshot, then transformed into a Falcon 9 analytical dataset.
- EDA identified differences across launch sites, orbit types, payload mass and launch year.
- SQL added grouped, ranked and aggregate views of launch and landing outcomes.
- Folium added geographic context around the four launch sites.
- Four classifiers were tuned with 10-fold cross-validation; the Decision Tree had the highest validation score (87.68%).

Source: supplied IBM Skills Network / Coursera lab notebooks

Introduction

Business question and analytical framing

WHY THIS MATTERS

Landing success drives first-stage reusability.

- The lab frames landing prediction as a classification problem.
- The target is the binary Class label created from landing outcomes.
- The project links descriptive analytics with a supervised predictive workflow.

PROJECT FLOW

01 **Collect**

↓

02 **Wrangle**

↓

03 **Explore**

↓

04 **Model**

↓

05 **Interpret**

Question: can available launch, payload, orbit and booster features predict first-stage landing success?

Source: supplied IBM Skills Network / Coursera lab notebooks

Data Collection Methodology

Two complementary acquisition paths.

SPACE X API

- GET request to the SpaceX launches endpoint.
- API lookups enrich records with booster version, launch-site coordinates, payload mass/orbit and core/landing attributes.
- json_normalize and targeted filtering produce the analytical launch table.
- Rows with multiple cores/payloads are excluded for the Falcon 9 modeling workflow.

WIKIPEDIA WEB SCRAPING

- A fixed 9 June 2021 Wikipedia snapshot was requested.
- BeautifulSoup identifies the launch tables and parses HTML rows.
- Column names are extracted from <th> headers.
- The scraper builds a Pandas DataFrame containing flight, date, site, payload, orbit, customer and launch-outcome fields.

Source: supplied IBM Skills Network / Coursera lab notebooks

Data Wrangling Methodology

From raw launch records to a model matrix.

TRANSFORMATION PIPELINE

Raw records	API / scraped fields
Falcon 9 filter	Remove Falcon 1
Clean features	Dates, payloads, coordinates
Landing label	Class = 1 successful; 0 otherwise
Model matrix	Categorical encoding + numeric casting

DATA QUALITY

28.9%
LandingPad missing

90
final analytical rows

The notebook reports no missing values for the core analytical columns except LandingPad.

Source: supplied IBM Skills Network / Coursera lab notebooks

EDA & Interactive Visual Analytics Methodology

Static EDA answers 'what varies?'; Folium answers 'where?'.

EDA

- Compare flight number with launch site and orbit.
- Compare payload mass with launch site and orbit.
- Compute success rate by orbit.
- Track annual success rate.
- Create dummy variables for categorical predictors.

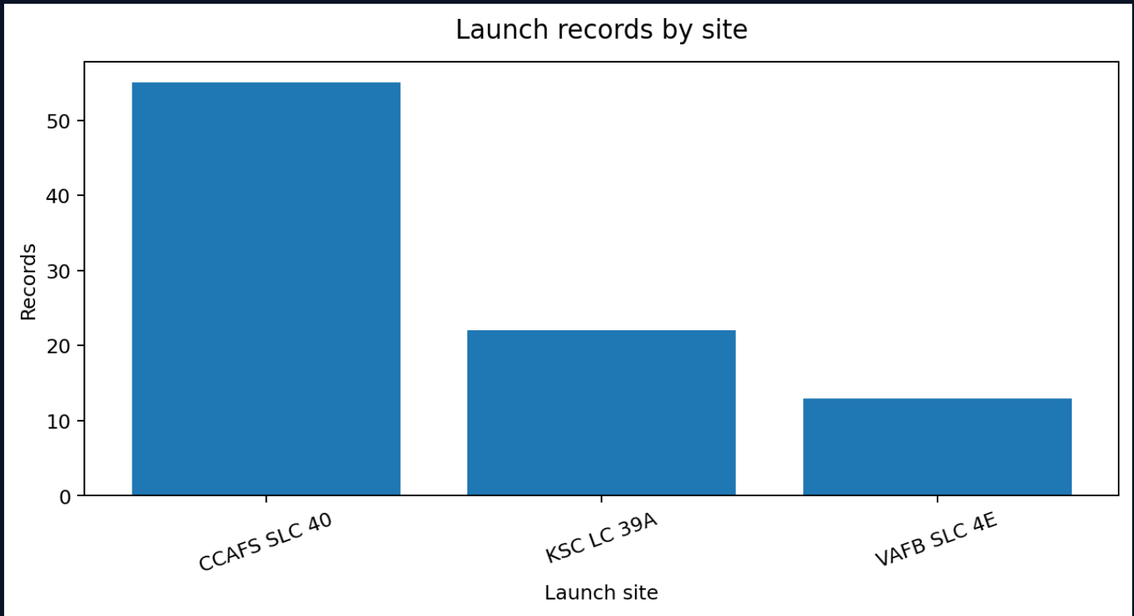
FOLIUM

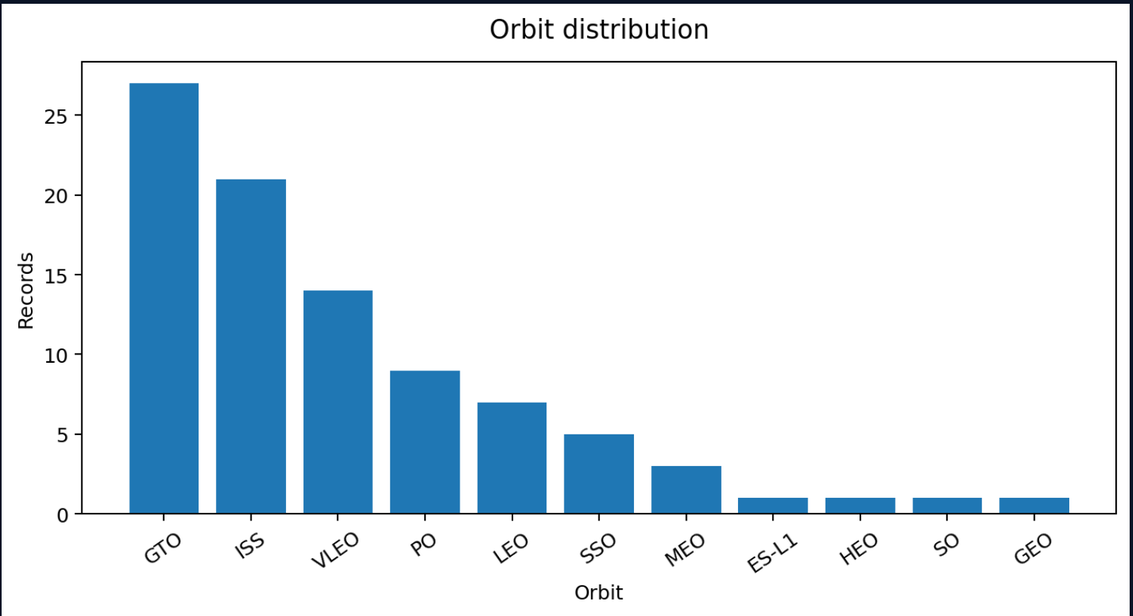
- Plot four launch sites from latitude/longitude.
- Use markers and clusters for successful vs failed launches.
- Add MousePosition to inspect nearby infrastructure.
- Calculate approximate distances.
- Reason about coastline, railways, highways and cities.

Source: supplied IBM Skills Network / Coursera lab notebooks

EDA Results I — Where and What Was Launched

Site and orbit concentration.





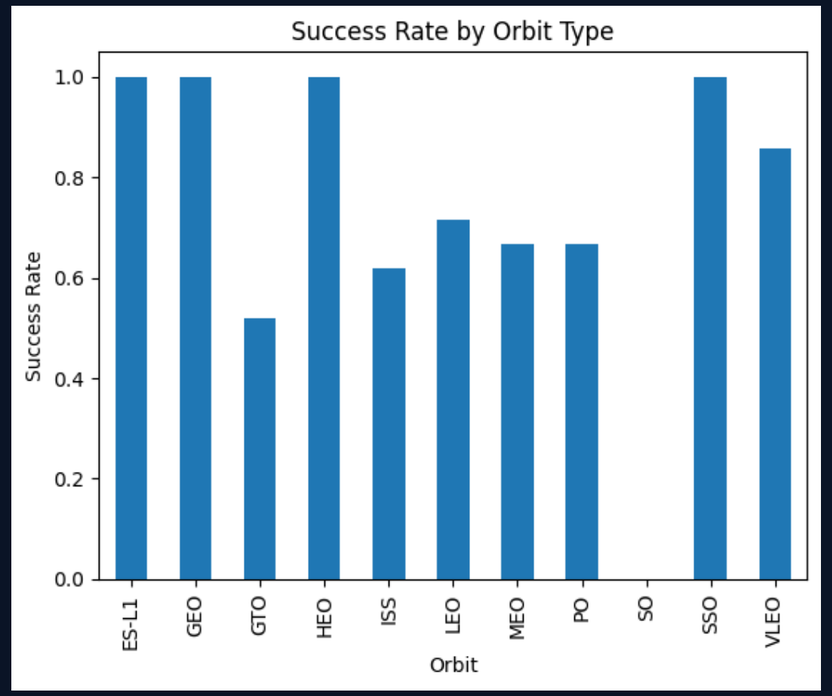
CCAFS SLC 40 = 55/90 records; KSC LC-39A = 22; VAFB SLC-4E = 13.

GTO = 27; ISS = 21; VLEO = 14; PO = 9; LEO = 7; SSO = 5; MEO = 3.

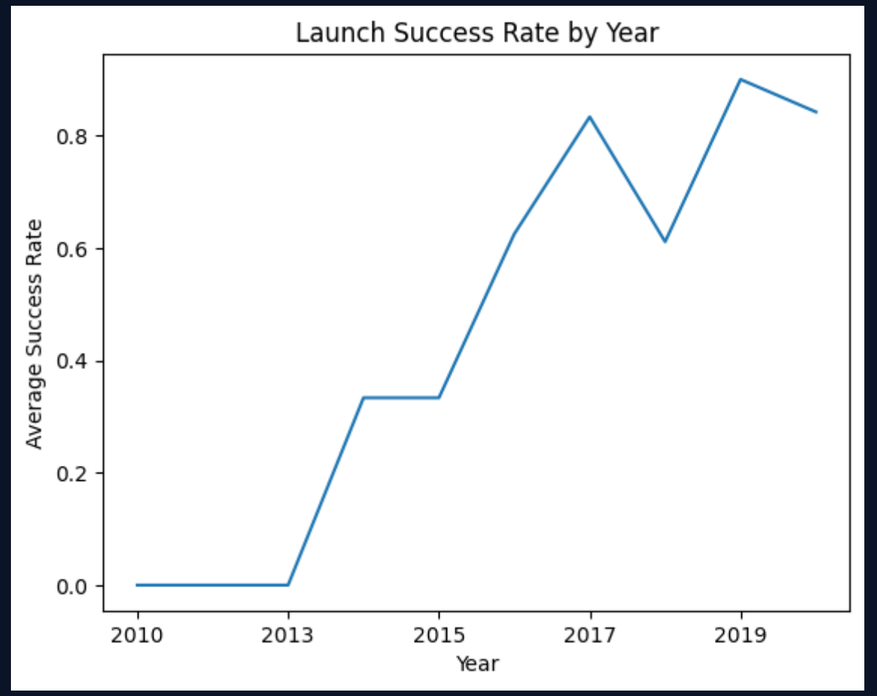
Source: supplied IBM Skills Network / Coursera lab notebooks

EDA Results II — Success Patterns Over Time and Orbit

Supplied notebook visualizations.



Orbit-level success rates vary substantially; several low-frequency categories reach 1.0.

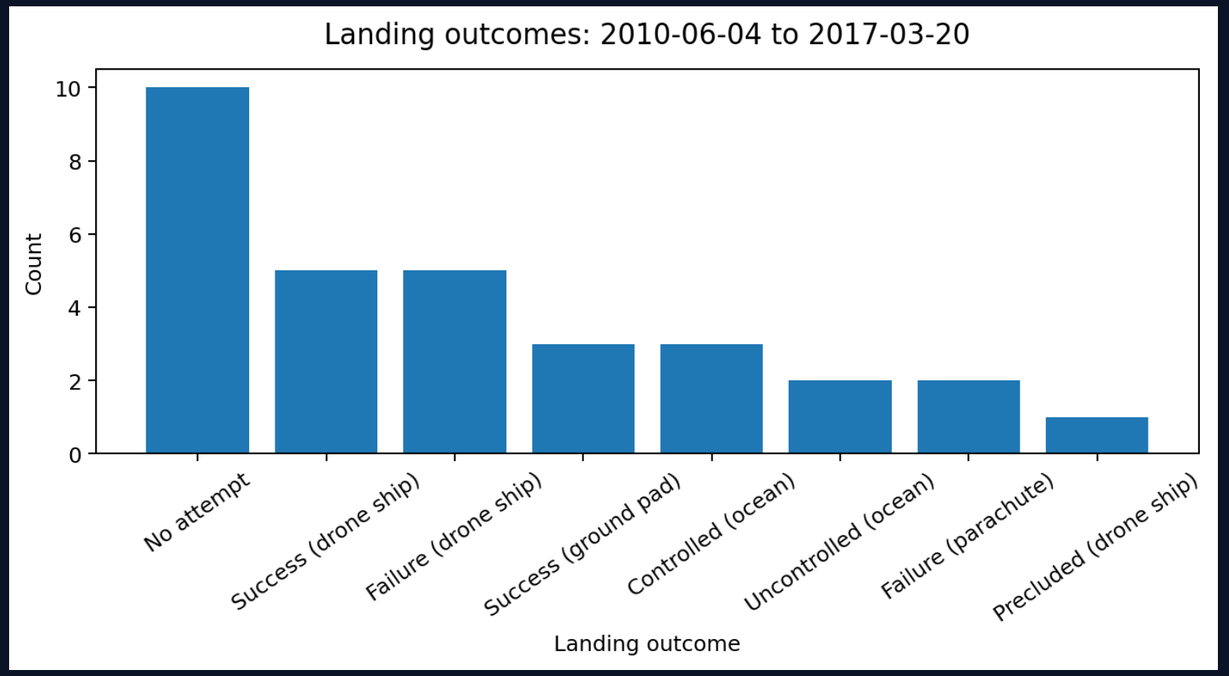


Annual average success rises strongly over the later years shown, with year-to-year variation.

Source: supplied IBM Skills Network / Coursera lab notebooks

EDA with SQL — Methodology & Results

SQLite: filtering, aggregation, subqueries and ranking.



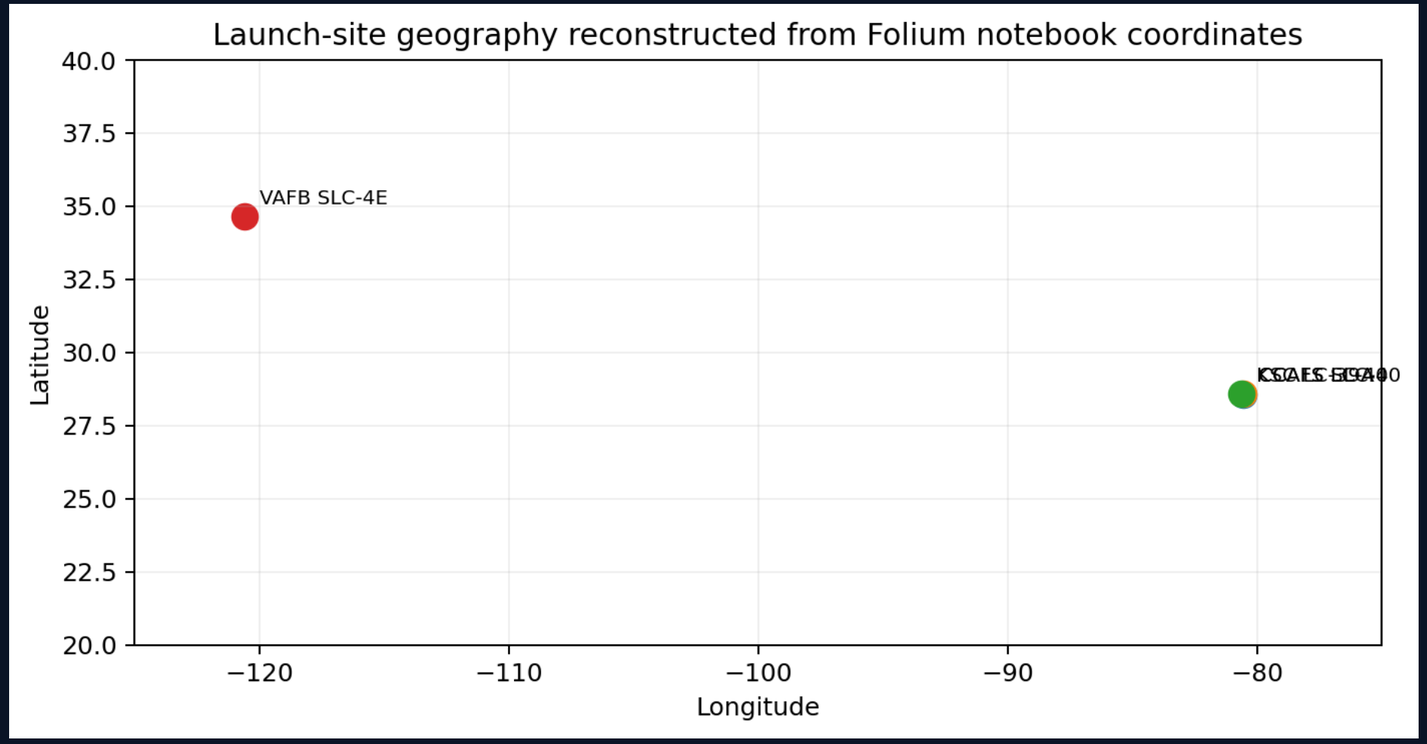
SELECTED FINDINGS

- 101 rows loaded into the SQL table.
- Four unique launch sites.
- Mission outcomes: 98 Success plus three other outcome variants.
- Maximum payload mass: 15,600 kg.
- 'No attempt' ranks first for the specified 2010–2017 landing-outcome window.

Source: supplied IBM Skills Network / Coursera lab notebooks

Interactive Map Results — Folium

Geographic analysis based on coordinates shown in the supplied Folium notebook.



MAP-BASED OBSERVATIONS

- Three sites cluster around Cape Canaveral / Kennedy Space Center.
- VAFB SLC-4E is on the California coast.
- All sites are north of the Equator at relatively low latitudes.
- The notebook adds marker clustering, MousePosition and distance lines.

Source: supplied Folium notebook; static geography reconstructed from its displayed coordinates.

Plotly Dash Dashboard — Submission Gap

No Dash notebook/output was included in the supplied project files.

WHAT IS PRESENT

- API collection, web scraping, wrangling, SQL, EDA, Folium and classification are supplied.
- No Plotly Dash implementation or rendered dashboard is present.
- An actual Dash result cannot be represented as completed source work without inventing evidence.

RECOMMENDED DASH LAYOUT

90
launch records

66.7%
landing success

55
CCAFS SLC 40

GTO
most frequent orbit

Filters: site • orbit • payload • booster • year

Views: success rate • launch trend • payload • model output

Source: supplied IBM Skills Network / Coursera lab notebooks

Predictive Analysis Methodology

Classification pipeline.

1

Target

Class

2

Standardize

StandardScaler

3

Split

80 / 20

4

Tune

GridSearchCV, cv=10

5

Evaluate

Validation + test

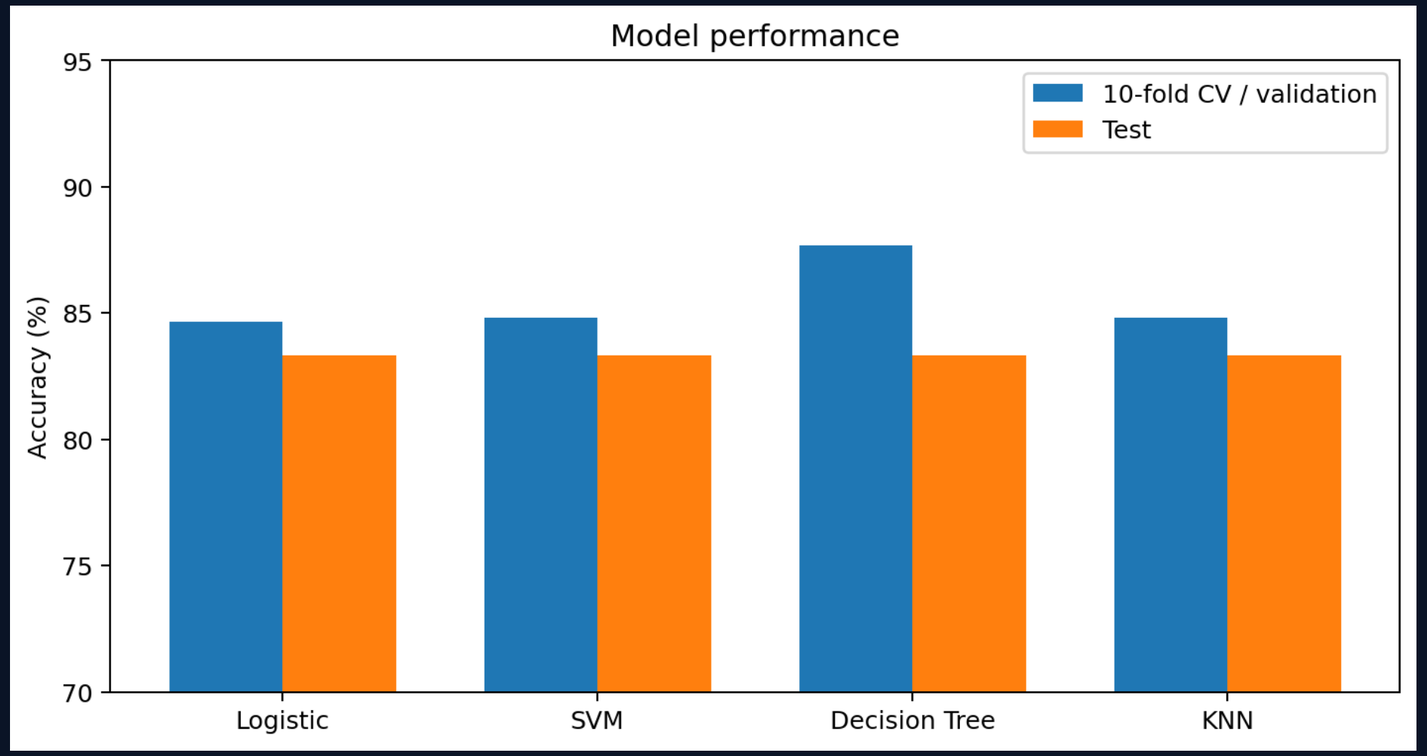
Models: Logistic Regression • SVM • Decision Tree • KNN

SVM search included linear, RBF, polynomial and sigmoid kernels.

Source: supplied IBM Skills Network / Coursera lab notebooks

Predictive Analysis Results — Model Comparison

Validation differs; held-out test accuracy is identical.



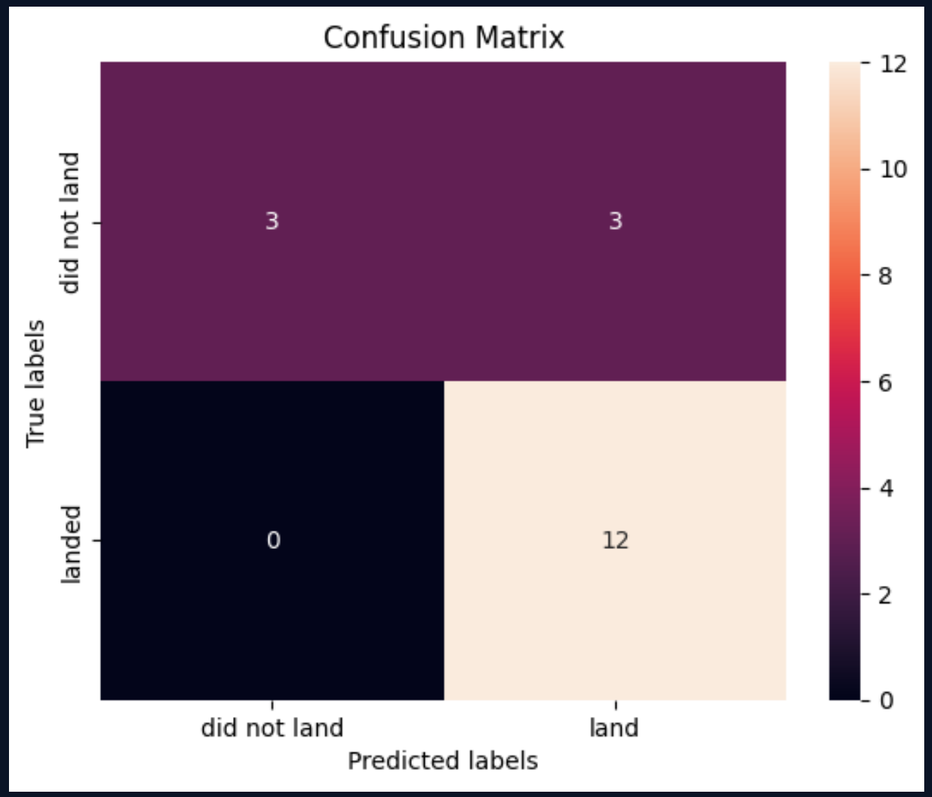
TUNED RESULTS

- Decision Tree: 87.68% CV.
- SVM (sigmoid): 84.82% CV.
- KNN: 84.82% CV.
- Logistic Regression: 84.64% CV.
- All four: 83.33% test accuracy on 18 samples.

Source: supplied IBM Skills Network / Coursera lab notebooks

Predictive Diagnostics — Confusion Matrix

The supplied notebook documents the same test matrix for the evaluated classifiers.



12
true positives

3
false positives

0
false negatives

3
true negatives

Documented error pattern: false positives — cases predicted as landed when the true label was 'did not land'.

Source: supplied IBM Skills Network / Coursera lab notebooks

Innovative Insights

Cross-module synthesis; descriptive rather than causal.

SITE CONCENTRATION

55/90 records = 61.1% at CCAFS SLC 40.

TARGET BALANCE

Class mean = 0.6667; successful landings are the majority class.

TIME EFFECT

Annual success improves markedly in the later EDA years.

MODEL STABILITY

All four tuned models reach 83.33% test accuracy.

These insights are directly derived from the supplied notebook outputs.

Conclusion & Submission Checklist

Final steps before submission.

CONCLUSION

- The notebooks support a collection → wrangling → EDA → SQL → Folium → ML narrative.
- The Decision Tree has the highest CV score (87.68%).
- The SVM's best kernel is sigmoid (84.82% CV).
- All four models reach 83.33% test accuracy.

BEFORE SUBMISSION

- Replace the GitHub placeholder with the real repository URL.
- Insert the actual Plotly Dash result if that lab exists outside the supplied files.
- Submit the PDF required by the course.

GitHub repository:

PASTE YOUR LINK HERE